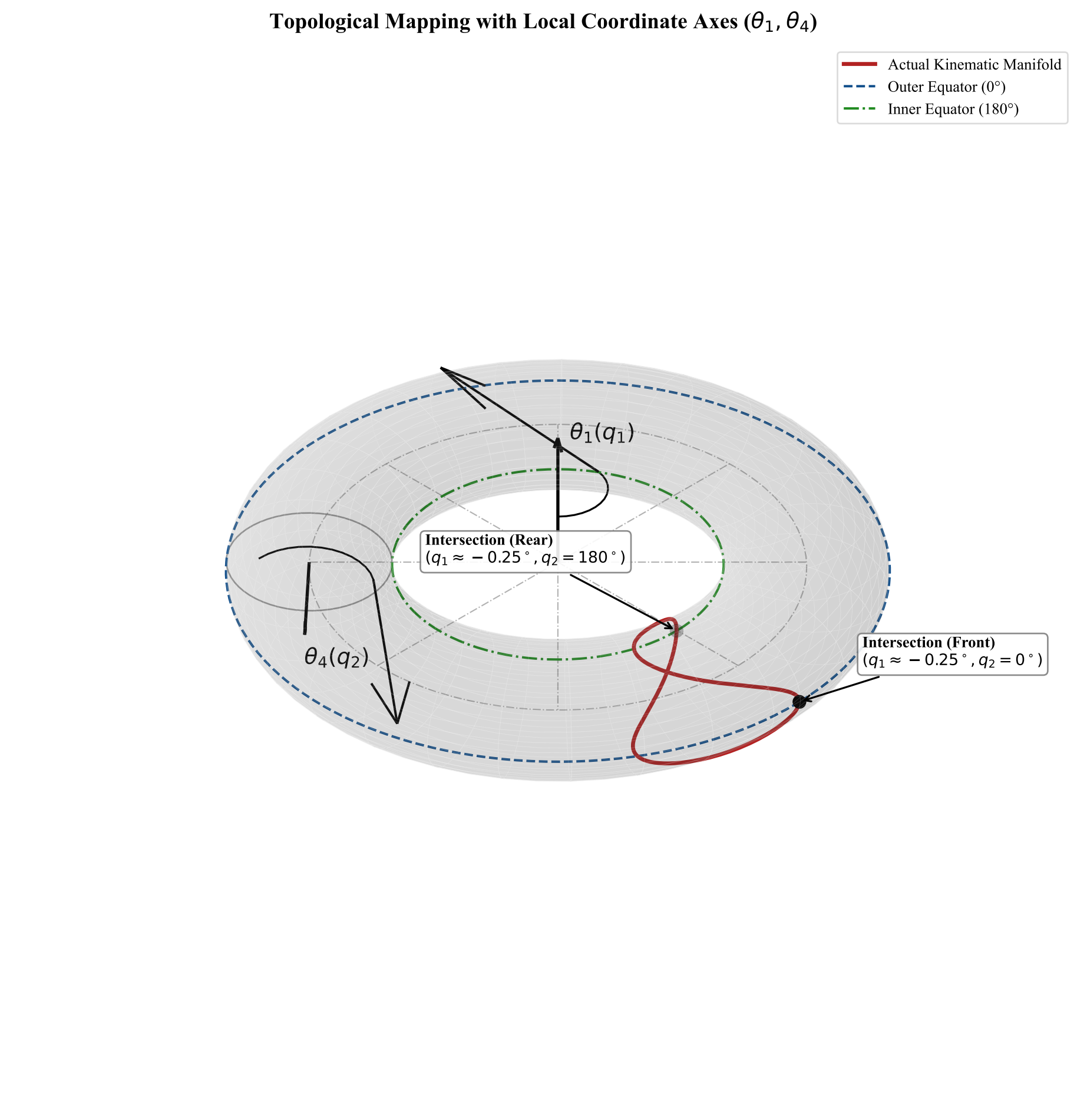

Topological Mapping with Local Coordinate Axes (θ_1, θ_4)
Actual Kinematic Manifold
Outer Equator (0°)
Inner Equator (180°)
$\theta_1(q_1)$
Intersection (Rear)
($q_1 \approx -0.25^\circ, q_2 = 180^\circ$)
$\theta_4(q_2)$
Intersection (Front)
($q_1 \approx -0.25^\circ, q_2 = 0^\circ$)